Linchuan Wei

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Research interests

Mixed-integer nonlinear and linear programming with applications in statistical learning, causal inference and logistics

Education

Northwestern University — Evanston, IL
Ph.D. in Operations Research — September 2018 – Present
- Ph.D. Advisor: Simge Küçükyavuz

University of Waterloo — Waterloo, Ontario
B.S. in Honours Mathematics — August 2018

Honors

Terminal Year Fellowship, Northwestern University — 2022
Nemhauser Best Paper Award, Northwestern University — 2021
Dean's Honour List, University of Waterloo — 2014-2018

Publications

Consistent second-Order conic integer programming for learning Bayesian networks
Simge Küçükyavuz, Ali Shojaie, Hasan Manzour, Linchuan Wei, Hao-Hsiang Wu.
Journal of Machine Learning Research, conditional accept, 2023.

On the convex hull of convex quadratic optimization problems with indicators
Linchuan Wei, Alper Atamtürk , Andrés Gómez, Simge Küçükyavuz.
Mathematical Programming, forthcoming, 2023.

Ideal formulations for constrained convex optimization problems with indicator variables
Linchuan Wei, Andrés Gómez, Simge Küçükyavuz.
Mathematical Programming, 192(1-2), 57-88, 2022.

On the convexification of constrained quadratic optimization problems with indicator variables
Linchuan Wei, Andrés Gómez, Simge Küçükyavuz.
International Conference on Integer Programming and Combinatorial Optimization, IPCO 2020, Lecture Notes in Computer Science(), vol 12125. Springer, Cham.

Robust principal component analysis using facial reduction
Shiqian Ma, Fei Wang, Linchuan Wei, Henry Wolkowicz.
Optimization and Engineering, 21, 1195–1219, 2020.

Teaching

Teaching assistant IEMS 313: Foundations of Optimization, Fall 2022 - Northwestern University

Grader IEMS 450-1: Mathematical Optimization I, Fall 2021 - Northwestern University

Guest lecturer and grader IEMS 457: Integer Programming, Winter 2021, Spring 2023 - Northwestern University

Talks

On the convex hull of convex quadratic optimization problems with indicators

Mixed-integer Programming Workshop May, 2023
INFORMS Annual Meeting October, 2022
International Conference on Continuous Optimization July, 2022

Ideal formulations for constrained convex optimization problems with indicator variables

Mixed-integer Programming Workshop May, 2021
INFORMS Annual Meeting November, 2020

On the convexification of constrained quadratic optimization problems with indicator variables

Conference on Integer Programming and Combinatorial Optimization June, 2020

Service and outreach

Referee: Mathematical Programming (2021-2023)

Co-founder: Integer programming and combinatorial optimization reading group (2019-2021)